



Society of Petroleum Engineers

SPE-230142-MS

Automating Process Safety Barrier Management Using AI: A Case Study on GCC Onshore COMAH Site

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This paper presents the development and implementation of an artificial intelligence (AI) solution aimed at automating the identification and verification of process safety barriers and generating performance standards. The goal is to enhance process safety management and operational efficiency in the energy industry. Our approach involved developing an AI system that leverages Retrieval-Augmented Generation (RAG) and the latest Large Language Models (LLMs) to identify Health, Safety, Environment, and Critical Equipment Systems (HSECES) equipment based on inclusion rules defined in primary engineering deliverables, HSECES management guidelines, Bow Tie, and COMAH reports. The AI system extracts and populates the HSECES register at the equipment level, referencing source documents. It then generates relevant performance standards, compares them with existing standards, and provides recommendations for improvement. The AI verifies implementation in the computerized maintenance management system (CMMS), reviews completed work orders, and highlights discrepancies and areas for improvement via an integrated dashboard. The system also performs reliability analysis to identify trends and maintenance optimization opportunities.

The incorporation of AI significantly reduced the required engineering effort and is scalable across existing assets and new sites. The work resulted in substantial improvements in the accuracy and efficiency of HSECES identification and performance standard generation. The AI successfully populated the HSECES register, ensuring accurate equipment identification and allocation of relevant barriers, verified by engineering review. The performance standards generated were comprehensive and aligned with site operator requirements and industry standards. Verification in CMMS ensured accurate task list development and implementation. Using AI to review completed work orders identified and addressed discrepancies, leading to improved compliance and practices. Reliability analysis highlighted areas for performance improvement and optimization, contributing to enhanced plant uptime and reduced engineering effort. The use of RAG and advanced LLMs facilitated real-time, contextually relevant data processing, ensuring robust control of major accident hazards.

The case study demonstrates the potential of AI to transform safety and maintenance processes, offering valuable insights and advancements in the field. The automation of these processes reduces incidents,

improves plant uptime, and reduces engineering effort by over 90%, setting a new benchmark for AI-driven safety management in the energy industry.

Introduction

The oil and gas sector remains one of the most technically complex and safety-critical industries globally, operating under conditions where the margins for error are exceptionally small. Processes such as hydrocarbon extraction, refining, and transportation involve high pressures, flammable substances, and intricate mechanical and control systems, where even minor deviations can precipitate catastrophic events (CCPS 2023). Consequently, process safety is not merely a regulatory obligation but a strategic priority underpinning operational continuity, environmental stewardship, and the protection of human life (API 2019).

Regulatory regimes such as the Control of Major Accident Hazards (COMAH) Regulations in the United Kingdom, the Seveso III Directive in the European Union, and OSHA's Process Safety Management (PSM) standard in the United States impose stringent requirements on hazard identification, risk assessment, and the maintenance of critical safety barriers (HSE 2015; OSHA 2019). At the heart of these frameworks lies the effective management of Health, Safety, Environment, and Critical Equipment Systems (HSECES) - the combination of engineered controls, procedural safeguards, and monitoring systems designed to prevent and mitigate major accident hazards (MAH).

Traditional approaches to HSECES management have relied heavily on manual data gathering, paper-based documentation, and periodic human-led inspections. While these approaches have been instrumental in improving safety performance over past decades, they tend to be reactive rather than proactive. Critical safety insights often emerge only after anomalies have been detected, incident investigations have been completed, or regulatory audits have identified deficiencies (CCPS 2021). Furthermore, operational variability across assets can result in inconsistent application of safety practices and knowledge silos that hinder organizational learning.

Recent advances in digital technology, and Artificial Intelligence (AI) in particular, have created opportunities to address these challenges. AI's ability to process, synthesize, and interpret vast quantities of heterogeneous data at machine speed positions it as a transformative enabler in the safety domain (World Economic Forum 2022). Among emerging techniques, Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) offer distinct advantages for industrial safety applications.

RAG architectures integrate an LLM with a domain-specific information retrieval system, allowing the model to access up-to-date, context-relevant documents from approved repositories before generating a response (Lewis et al. 2020). This reduces the risk of factual inaccuracies, a critical consideration in safety-critical contexts, while enabling highly contextualized decision support. LLMs, meanwhile, excel in natural language understanding and generation, enabling them to automate labor-intensive textual tasks such as compiling regulatory reports, summarizing maintenance logs, generating inspection checklists, and answering natural language engineering queries (Bommasani et al. 2021).

The integration of RAG and LLMs into process safety management offers multiple potential benefits:

- Enhanced hazard awareness through real-time retrieval and synthesis of operational data, incident histories, and regulatory requirements.
- Reduced engineering workload by automating assignment of categories (such as HSECES criticality) from documentation, compliance reporting, and cross-referencing of safety cases with live asset data.
- Improved consistency in safety practices across geographically dispersed assets by standardizing interpretation of procedures and regulations.
- Proactive risk management through AI-assisted trend analysis and early detection of deviations in barrier performance indicators.

Beyond operational efficiency, AI-enabled safety management supports the industry's broader energy transition objectives. As operators integrate renewable generation, electrification, and carbon capture technologies into their portfolios, the complexity of asset configurations will increase (IEA 2023). AI can serve as a unifying digital layer, harmonizing safety oversight across legacy and emerging technologies, and enabling operators to maintain high safety standards while pursuing decarbonization and cost optimization.

This paper presents a pioneering case study of an AI-driven process safety framework implemented at a Gulf Cooperation Council (GCC) onshore COMAH site. The system integrates RAG-enabled LLMs into existing safety management structures, automating key compliance tasks, improving data traceability, and providing engineers with immediate, contextually accurate information for safety decision-making. The deployment demonstrates measurable reductions in manual engineering effort, improvements in the consistency of HSE management, and enhanced responsiveness to potential deviations from safe operating envelopes.

By situating the discussion within both the academic literature and real-world operational experience, this paper contributes to the growing body of knowledge on AI-enabled safety systems and offers a replicable model for high-hazard industries seeking to transition from reactive, document-heavy safety management to proactive, insight and data-driven risk control.

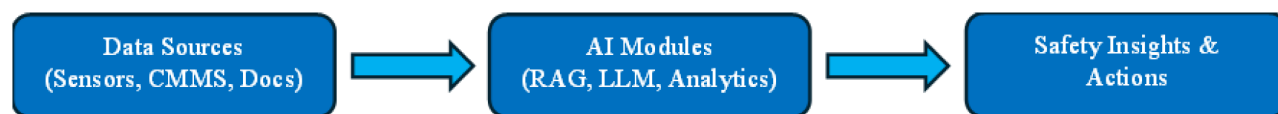


Figure 1—AI Integration in Process Safety Management

Background and Literature Review

Process Safety Management (PSM) refers to a structured framework of practices, systems, and processes designed to prevent the occurrence of major accident hazards in high-risk industries such as oil and gas, chemicals, and petrochemicals. At its core, PSM integrates engineering, operational, and management activities to ensure that hazardous processes remain within safe operating limits throughout their lifecycle (CCPS 2021).

Key components of PSM include hazard identification, risk assessment, barrier management, and performance standard verification. Hazard identification involves systematically determining the full range of credible process deviations that could lead to hazardous events, often using methodologies such as Hazard and Operability (HAZOP) studies and Layer of Protection Analysis (LOPA) (Kletz 2009). Risk assessment evaluates the likelihood and consequences of identified hazards, informing decisions about which safeguards are necessary and proportionate. Barrier management ensures that both engineered and procedural barriers are in place, functional, and capable of preventing escalation. Performance standard verification then provides assurance that each barrier meets the design, operational, and maintenance criteria necessary for it to remain effective over time (HSE 2015).

Traditionally, these PSM tasks require extensive manual review of engineering documentation, Bow Tie analyses, COMAH safety reports, maintenance logs, and incident records. Engineers and safety professionals typically spend substantial time consolidating information from disparate data sources, a process prone to inconsistency, human error, and latency in risk detection (Aderamo et al. 2024). For example, the verification of HSE performance standards often involves laborious cross-referencing between technical specifications, inspection reports, and maintenance histories.



Figure 2—Bow Tie Diagram for Hazard and Barrier Management

Recent studies have begun to highlight the limitations of purely manual PSM approaches and the potential of Artificial Intelligence (AI) to augment or transform these processes. [Balasubramanian et al. \(2024\)](#) demonstrated how natural language processing techniques could be applied to large repositories of safety documentation to automatically extract hazard scenarios and map them to control measures, significantly reducing the time required for hazard identification. [Aderamo et al. \(2024\)](#) explored predictive analytics models trained on maintenance and incident data, showing their capability to forecast barrier degradation before functional failures occur.

The broader literature on AI in safety-critical industries underscores several potential areas of impact. AI has been applied in automated hazard identification using computer vision for facility inspections ([Zhou et al. 2022](#)), real-time risk assessment via machine learning models trained on process sensor data ([Sun et al. 2021](#)), and reliability analysis for maintenance optimization ([Jardine, Lin, and Banjevic 2006](#)). Machine learning models, including large language models (LLMs), are particularly promising for extracting structured safety insights from unstructured sources such as incident narratives, inspection notes, and regulatory guidance ([Bommasani et al. 2021](#)).

The objectives of this study are fourfold:

1. Automate the identification and verification of HSECES equipment.
2. Generate and validate performance standards using AI.
3. Integrate AI into Computerized Maintenance Management Systems (CMMS).
4. Perform reliability analysis for maintenance optimization.

A further strategic goal is to demonstrate scalability across assets, ensuring that the AI framework can be deployed not only at the pilot COMAH site but also across diverse facilities within an operator's portfolio. This addresses a recurring challenge in PSM: the difficulty of achieving consistent application of safety standards across multiple sites with differing equipment configurations, operational histories, and local regulatory requirements ([CCPS 2023](#)).

While the application of AI in PSM is still emerging, there is growing evidence from both the process safety literature and the broader industrial safety domain that intelligent automation can transition safety management from a reactive, document-heavy discipline to a proactive, insight-driven function. This positions AI not as a replacement for engineering judgment, but as an enabler that amplifies the reach, speed, and precision of human expertise.

Methodology

This work was conducted as a pilot on a single COMAH site to test feasibility, quantify effort reduction, and identify integration constraints. All AI outputs were treated as first-pass aids to the engineering team and did not replace formal risk assessment activities such as HAZOP, LOPA, or the Bow Tie process that identifies

HSECES. Integration with the live CMMS was assessed on a desktop basis only; no production write-backs or automated scheduling changes were performed during the pilot. The objective was to evaluate whether retrieval-augmented language models could accelerate document interrogation and drafting while keeping engineering judgement as the decision authority.

Document corpus and curation

The corpus comprised approved site deliverables, including HSECES management guidelines, Bow Tie analyses, COMAH submissions, selected equipment registers, and maintenance task lists. Documents were placed in a controlled repository with versioning and access logging. Only in-scope documents were indexed to preserve traceability; superseded or ambiguous sources were flagged for reviewer decision. Where inputs existed only as scanned images, optical character recognition was applied and confidence scores recorded. Low-confidence extracts were routed to manual abstraction by engineers.

AI architecture and retrieval

A retrieval-augmented generation (RAG) workflow was used to ensure that model outputs remained grounded in the curated corpus. A semantic retriever selected passages from the repository and supplied them to a large language model, which produced candidate extractions and drafts with explicit source pointers. The system was configured to return "needs review" when the retriever could not supply sufficient evidence, rather than force an answer. Prompt templates and retrieval filters were fixed for the duration of the pilot to support repeatability.

First-pass extraction of candidate HSECES

Using the RAG workflow, the model proposed candidate HSECES items and attributes for a dynamic register, together with citations to the underlying documents. The intention was speed, not authority. Candidates were then passed to the engineering review described below. All insertions, deletions, and edits were recorded with rationale and preserved as an auditable trail that could be challenged during quality assurance or by regulators.

Human-in-the-loop engineering review

Each AI pass was followed by a structured review by at least two engineers. Reviewers confirmed scope, cross-checked inclusions against Bow Tie diagrams and COMAH basis of safety, corrected misassignments, and approved accepted items for downstream use. Disagreements were resolved in a short adjudication step and the final decision, with links to sources, was captured in the register. This human-in-the-loop stage was the primary control that kept the pilot aligned with engineering practice and prevented over-reliance on model suggestions.

Drafting and comparison of performance standards

For the reviewed HSECES items, the model generated draft text for performance standards, including purpose, functional requirements, assurance tasks, and impairment response. Drafts were automatically compared with existing site and corporate standards to surface potential gaps, overlaps, and inconsistencies. Engineers then edited or rejected suggestions as required. The AI role remained assistive; ownership of the final content rested with the engineering team.

Barrier verification support

To assist barrier verification, the system retrieved and assembled the most relevant clauses, drawings, and narratives tied to the barriers in the Bow Tie analyses. The pilot did not attempt live assessment of barrier health and made no claims about operational status. Instead, it accelerated access to the right evidence so that reviewers could make faster, better-informed judgements during routine checks.

CMMS alignment and desktop review

Reviewed HSECEs items and draft assurance requirements were compared against exported maintenance task lists from the CMMS. The analysis highlighted potential issues such as assurance tasks attached to non-HSECEs equipment, missing procedures, and frequency mismatches. Findings were presented as recommendations for engineering consideration. No automatic task creation, deletion, or rescheduling occurred in the live CMMS.

Exploratory reliability and trend analysis

Historical maintenance records and failure logs were analysed to surface recurrent failure modes and candidates for maintenance optimisation. Results were framed as prioritised leads for further engineering review rather than prescriptive plans. Any change to maintenance strategy remained subject to the site's management-of-change process.

Model validation and benchmarking

Model performance was benchmarked against engineering judgement using a labelled hold-out set. Two engineers independently assigned labels at the item level for the classification task HSECEs versus Not HSECEs across a stratified sample of equipment classes and document types; disagreements were resolved by a third reviewer to form the reference set. With prompts and retrieval settings frozen, we computed precision, recall, F1, and constructed a confusion matrix with axes Prediction (HSECEs, Not HSECEs) and Actual (HSECEs, Not HSECEs) a 28×28 matrix. Barrier-specific matrices and error patterns were computed to build confidence and change parameters; the methodology here is limited to the evaluation design and metrics. Regardless of scores, engineering review remained mandatory before any output was accepted.

Quality assurance and risk mitigation

The methodology recognised the risks of hallucination, corpus gaps, and limitations in processing non-digitised deliverables. Hallucination risk was mitigated by constraining the model to cite only from the curated repository and by requiring source pointers for all critical claims; outputs without auditable sources were returned as needs review. Data quality and staleness were managed through version control and by flagging items tied to superseded or conflicting documents for explicit reviewer attention. For scanned P&IDs and other drawings, optical character recognition was used only to support text search; diagram interpretation remained a reviewer task, and low-confidence OCR was routed to manual abstraction. Governance controls included change logs for all edits, reviewer sign-off with rationale, and the explicit exclusion of live CMMS write-backs. Any future deployment would require formal management-of-change, cybersecurity assessment, and training to preserve these assurance layers.

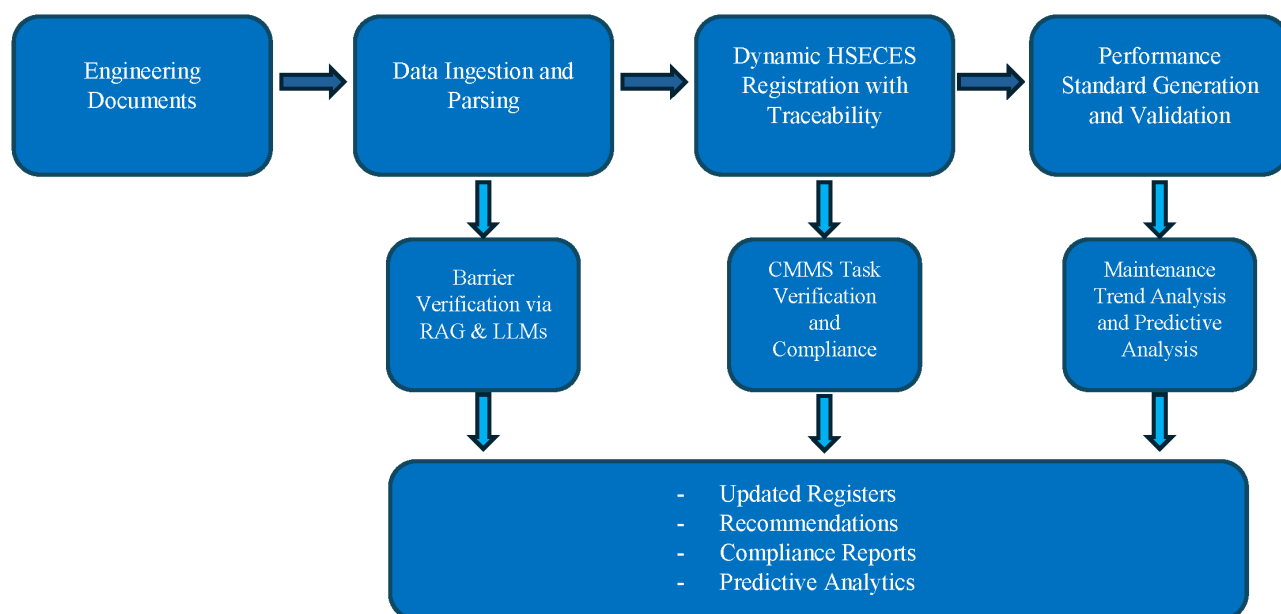


Figure 3—AI System Architecture for Process Safety Management

Results and Discussion

This study set out to provide a solution which is scalable across existing assets and new sites, making it a viable model for enterprise-wide deployment. Additionally, there were four key aims:

1. Automate the identification and verification of HSECES equipment.
 - The AI system achieved over 90% accuracy in identifying HSECES equipment, verified by engineering review.
 - The quality and layout of the source engineering documentation directly impacted the accuracy and recall of the AI models for HSECES identification. For example, controls and instrumentation equipment which could be verified from an Instrument Index required significantly less engineering effort to ensure AI accuracy compared to systems which required use of engineering drawings (e.g. P&IDs). Improvements in the ability of AI models to read drawings will further enhance the abilities of this approach to reduce manual engineering effort.
 - On a number of occasions, the AI models were successful in identifying equipment which had been incorrectly assigned by Engineers when the Asset Register was built.
 - Improving the quality of how HSECES barriers, and their boundaries, were assigned and described in the source documentation would further enhance the quality of the AI output

Figure 4 shows the accuracy and recall results for three sample barriers. Note that for each barrier the accuracy of the AI models exceeded that of the existing manually derived HSECES assignments.
2. Generate and validate performance standards using AI.
 - Performance standards generated were comprehensive and aligned with site, parent organization and industry requirements.
 - Manual effort to create these was reduced by over 90%, freeing up engineering resources for more value-adding tasks
3. Integrate AI into Computerized Maintenance Management Systems (CMMS).
 - AI was successfully used to conduct an assessment of the following areas within the CMMS system:

- HSECES equipment has appropriate HSECES assurance tasks
 - HSECES maintenance has maintenance procedures
 - HSECES maintenance is only conducted on safety critical equipment
 - Performance standard tasks are addressed by the assigned maintenance
- This identified where there were risk areas for the asset, and also opportunities to release value by eliminating maintenance tasks where it was shown that HSECES assurance tasks may have been incorrectly allocated to non-HSECES equipment. This identified (Figure 5):
- 2,373 pieces of equipment has maintenance tasks which could be removed
 - 58 maintenance items had HSECES assurance tasks but no associated HSECES equipment
 - 223 assurance tasks where it was unclear if the requirements were being met by maintenance
- This study output is under review by the asset discipline engineering and technical authority teams at the time of writing however it is clear that the AI output has identified areas for improvement in performance and reduction in risk that was difficult for a human subject matter expert to identify.
4. Perform reliability analysis for maintenance optimization.

	CMMS		AI Output vs Client Engineer review		CMMS vs Client Engineer Review	
Unit 1	PC-03		PC-03		PC-03	
	Equipment Count	~200	Accuracy	87%	Accuracy	84%
			Recall	54%	Recall	29%
	DS-03		DS-03		DS-03	
	Equipment Count	~800	Accuracy	100%	Accuracy	98%
			Recall	100%	Recall	75%
Unit 2	PC-03		PC-03		PC-03	
	Equipment Count	~300	Accuracy	91%	Accuracy	91%
			Recall	75%	Recall	44%

Figure 4—Accuracy & Recall figures from Proof of Concept study

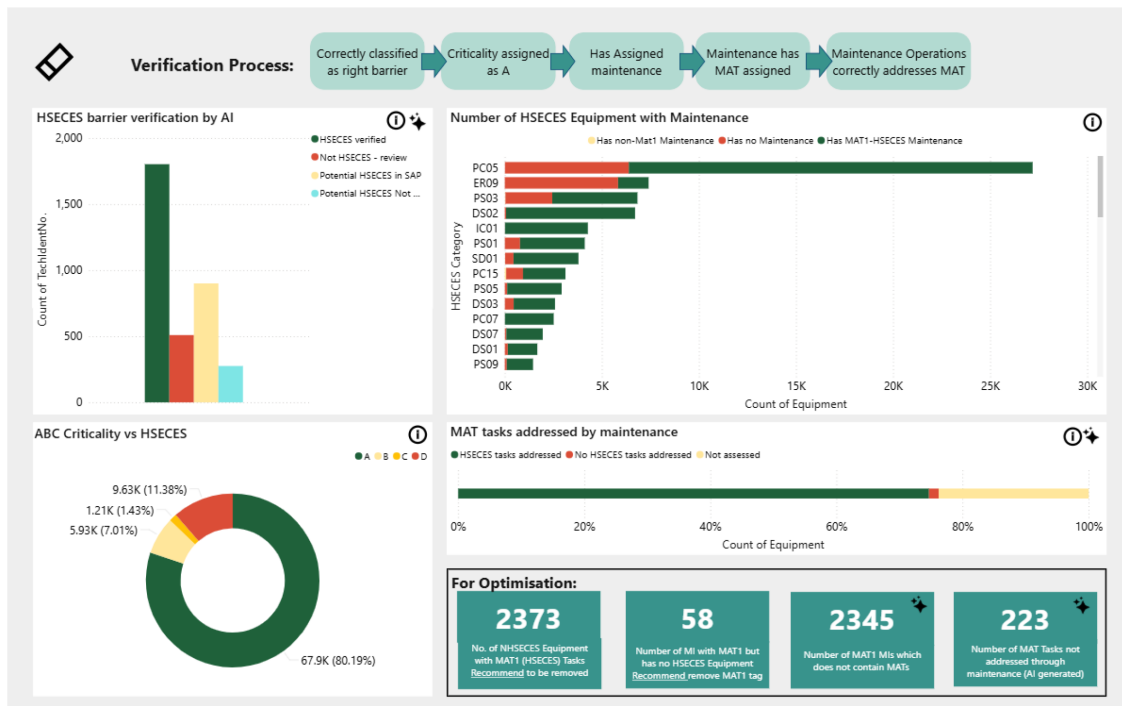


Figure 5—Extract from dashboard showing outputs of CMMS Verification process

By integrating AI into a maintenance optimization program, it has been shown (Stewart, 2025) that an overall time saving of 97% can be achieved in determining failure mode and understanding how to optimize maintenance tasks. Our findings in this case study showed a similar improvement in speed of analysis compared to manual efforts.

Over 83,000 maintenance hours were reviewed as part of this Pilot work and opportunities identified to remove up to 27,000 annual hours based upon the historic reliability of the equipment. The full outcome of the management of change for these opportunities is still underway at time of writing. Figure 6 shows an example of the insights generated as part of this process.

Plant	SME Name	Discipline	Cluster	SECE	Maint. Item	Tasklist Desc.	Min Population MTBF Y	5.07	(Blank)	(Blank)	(Blank)							
751 752 753 754	All	All	All	All	All	All												
Object Type Search Filters & Strainers Fire&Gas Detectors Firefighting E Heat Exchangers	Tasklist	Long Text					ObjType-FM	751										
FM Search Select all Abnormal Instrument Reading Abnormal Output High	2806-1	PM-6M-Flammable (H2)Gas Det-FGS HSE TASKS ----- Conduct Task Risk Assessment Obtain Permit To Work (PTW) Conduct Tool Box Talk Perform All Equipment Isolation and De-isolation, as required. Ensure All Permit To Work (PTW) conditions are met. Check Adherence To HSE Procedures Confirm All SIMOPS interfaces. PM-6M-Flammable (H2)Gas Det-FGS ----- Test and Calibrate as per Cause and Effect: Initiate High Alarm Value greater than or equal to 10 % LEL. Record value and update in work order.					Fire&Gas Detectors - Abnormal Instrument Reading Fire&Gas Detectors - Abnormal Output High Fire&Gas Detectors - Abnormal Output Low Fire&Gas Detectors - Erratic Output Fire&Gas Detectors - Failure to function Fire&Gas Detectors - No Output Fire&Gas Detectors - Unspecified Failure Mode	5.08 5.08 5.08 5.08 5.08 5.08 5.07										
Cluster	HSECS Category	Discipline	SECE	Tasklist	Tasklist Description	Maintenance Plan	Maintenance Plan Description	Maintenance Item	Maintenance Item Description	Functional Location	Diff. Ops.	Operation Short Text	MIN Pop. Frequency	MTBF	Annualised Hours	Operator A	Operator B	Operator C
48	DS03	Instrument	Yes	2817-1	PM-H2S GAS DET-POINT TYPE-FGS	400000412110	PM-0751-H2S-G201	125675	PM-H2S GAS DET-POINT TYPE-FGS	A-SG-SHAH-0751-FD00_G201	Yes	PM-2Y-H2S Detector-Point Type-FGS	24	5.01	0.50			
48	DS03	Instrument	Yes	2817-1	PM-H2S GAS DET-POINT TYPE-FGS	400000412110	PM-0751-H2S-G201	125675	PM-H2S GAS DET-POINT TYPE-FGS	A-SG-SHAH-0751-FD00_G201	Yes	PM-6M-H2S Detector-Point Type-FGS	6	5.01	1.20			
48	DS03	Instrument	Yes	2818-1	PM-H2S GAS DET-POINT TYPE-FGS	400000412111	PM-0751-H2S-G202	125676	PM-H2S GAS DET-POINT TYPE-FGS	A-SG-SHAH-0751-NB20_G202	Yes	PM-2Y-H2S Detector-Point Type-FGS	24	5.01	0.70			
48	DS03	Instrument	Yes	2818-1	PM-H2S GAS DET-POINT TYPE-FGS	400000412111	PM-0751-H2S-G202	125676	PM-H2S GAS DET-POINT TYPE-FGS	A-SG-SHAH-0751-NB20_G202	Yes	PM-6M-H2S Detector-Point Type-FGS	6	5.01	2.00			
17	DS02	Instrument	Yes	2844-1	PM-1Y-SMOKE DETECTOR-FGS	400000412116	PM-0751-NE-G202	125681	PM-SMOKE DETECTOR-FGS	A-SG-SHAH-0751-NB20_G202	No		12	5.01	19.40			
17	DS02	Instrument	Yes	2844-3	PM-1Y-SMOKE DETECTOR-FGS	400000412116	PM-0751-NE-G202	125682	PM-SMOKE DETECTOR-FGS	A-SG-SHAH-0751-NB20_G202	No		12	5.01	24.80			
17	DS02	Instrument	Yes	2844-2	PM-1Y-SMOKE DETECTOR-FGS	400000412116	PM-0751-NE-G202	125683	PM-SMOKE DETECTOR-FGS	A-SG-SHAH-0751-NB20_G202	No		12	5.01	26.60			
2	DS07	Instrument	Yes	2845-1	PM-1Y-MANUAL CALL POINT-FGS	400000412106	PM-0751-NS-G201	125669	PM-MANUAL CALL POINT-FGS	A-SG-SHAH-0751-FD00_G201	No		12	5.01	20.00			
2	DS07	Instrument	Yes	2846-1	PM-1Y-MANUAL CALL POINT (ES)-FGS	400000412107	PM-0751-NS-G202	125670	PM-MANUAL CALL POINT (ES)-FGS	A-SG-SHAH-0751-NB20_G202	No		12	5.01	11.60			
2	DS07	Instrument	Yes	2846-2	PM-1Y-MANUAL CALL POINT (SS)-FGS	400000412107	PM-0751-NS-G202	125671	PM-MANUAL CALL POINT (SS)-FGS	A-SG-SHAH-0751-NB20_G202	No		12	5.01	8.60			
16	DS02	Instrument	Yes	2980-1	PM-1Y-NEAR HEAT DETECTOR-FGS	400000412117	PM-0751-RECA-G202	125684	PM-NEAR HEAT DETECTOR-FGS	A-SG-SHAH-0751-FD00_G201	No		12	5.01	5.00			
16	DS02	Instrument	Yes	2979-2	PM-1Y-HEAT DETECTOR-FGS	400000412119	PM-0751-RE-G202	125685	PM-HEAT DETECTOR-FGS	A-SG-SHAH-0751-FD00_G201	No		12	5.01	27.20			

Figure 6—Maintenance optimization insights

Conclusion

This work shows that the application of AI in HSECES barrier assurance and maintenance optimization offers a significant step forward in asset management and major accident hazard risk management. This case study highlights how a scalable AI solution can demonstrate tangible value to the operators within asset-intensive industries.

AI performance depends on data quality. Future work will focus on improving document digitization, enhancing semantic tagging, and integrating real-time sensor data. Using frameworks like PIKE-RAG, the system can evolve by analyzing operation logs, fine-tuning models with expert feedback, and adapting to new regulations and site conditions. The framework can also be extended to environmental monitoring, emergency response planning, and asset integrity management.

This implementation represents novel AI-based framework for automating process safety barrier management. By integrating RAG and LLMs, the system enhances accuracy, reduces engineering effort, and improves compliance. The case study at a GCC COMAH site validates its effectiveness and scalability.

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